and

$$\begin{aligned}F\left(\frac{2\tau}{1-\tau}\right) &= F(-2+2/(1-\tau))\\ &= F(2/(1-\tau))\\ &= \left(\frac{1-\tau}{2}\right)^2 F\left(\frac{\tau-1}{2}\right) - 2\pi i\left(\frac{\tau-1}{2}\right).\end{aligned}$$

Hence,

$$F(1/2-1/2\tau) = \tau^2 F\left(\frac{\tau-1}{2}\right) - \frac{2\pi i 2\tau}{1-\tau} - 2\pi i\frac{(2\tau)^2}{(\tau-1)^2}\left(\frac{\tau-1}{2}\right).$$

But $F(2-2/\tau) = F(-2/\tau) = (\tau^2/4)F(\tau/2) - 2\pi i\tau/2$, and hence

$$\begin{aligned}E_2^*(1-1/\tau) &= F(1/2-1/2\tau) - 4F(2-2/\tau)\\ &= \tau^2\left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2)\right] - 2\pi i\left(\frac{2\tau}{1-\tau}+\frac{2\tau^2}{\tau-1}\right) + 4\pi i\tau\\ &= \tau^2\left[F\left(\frac{\tau-1}{2}\right) - F(\tau/2)\right].\end{aligned}$$

This proves (11). Then, the last property follows from it and the fact that

$$F(\tau) = \frac{\pi^2}{3} - 8\pi^2\sum_{k=1}^{\infty}\sigma_1(k)e^{2\pi ik\tau}.$$

Thus Proposition 3.8 is proved.

We can now conclude the proof of the four-squares theorem by considering the quotient $f(\tau) = E_2^*(\tau)/\theta(\tau)^4$, and applying Theorem 3.4, as in the two-squares theorem. Recall $\theta(\tau)^4 \to 1$ and $\theta(1-1/\tau)^4 \sim 16\tau^2 e^{\pi i\tau}$, as $\mathrm{Im}(\tau)\to\infty$. The result is that $f(\tau)$ is a constant, which equals $-\pi^2$ by Proposition 3.8. This completes the proof of the four-squares theorem.

## 4 Exercises

**1.** Prove that

$$\frac{(\Theta'(z|\tau))^2 - \Theta(z|\tau)\Theta''(z|\tau)}{\Theta(z|\tau)^2} = \wp_\tau(z-1/2-\tau/2) + c_\tau,$$

where $c_\tau$ can be expressed in terms of the first two derivatives of $\Theta(z|\tau)$, with respect to $z$, at $z = 1/2+\tau/2$. Compare this formula with the result in Exercise 5 in the previous chapter.

**2.** Consider the **Fibonacci numbers** $\{F_n\}_{n=0}^{\infty}$, defined by the two initial values $F_0 = 0$, $F_1 = 1$ and the recursion relation

$$F_n = F_{n-1} + F_{n-2} \qquad \text{for } n \geq 2.$$

(a) Consider the generating function $F(x) = \sum_{n=0}^{\infty} F_n x^n$ associated to $\{F_n\}$, and prove that

$$F(x) = x^2 F(x) + xF(x) + x$$

for all $x$ in a neighborhood of 0.

(b) Show that the polynomial $q(x) = 1 - x - x^2$ can be factored as

$$q(x) = (1 - \alpha x)(1 - \beta x),$$

where $\alpha$ and $\beta$ are the roots of the polynomial $p(x) = x^2 - x - 1$.

(c) Expand the expression for $F$ in partial fractions and obtain

$$F(x) = \frac{x}{1 - x - x^2} = \frac{x}{(1 - \alpha x)(1 - \beta x)} = \frac{A}{1 - \alpha x} + \frac{B}{1 - \beta x},$$

where $A = 1/(\alpha - \beta)$ and $B = 1/(\beta - \alpha)$.

(d) Conclude that $F_n = A\alpha^n + B\beta^n$ for $n \geq 0$. The two roots of $p$ are actually

$$\alpha = \frac{1 + \sqrt{5}}{2} \qquad \text{and} \qquad \beta = \frac{1 - \sqrt{5}}{2},$$

so that $A = 1/\sqrt{5}$ and $B = -1/\sqrt{5}$.

The number $1/\alpha = (\sqrt{5} - 1)/2$, which is known as the **golden mean**, satisfies the following property: given a line segment $[AC]$ of unit length (Figure 2), there exists a unique point $B$ on this segment so that the following proportion holds

$$\frac{AC}{AB} = \frac{AB}{BC}.$$

If $\ell = AB$, this reduces to the equation $\ell^2 + \ell - 1 = 0$, whose only positive solution is the golden mean. This ratio arises also in the construction of the regular pentagon. It has played a role in architecture and art, going back to the time of ancient Greece.

**3.** More generally, consider the difference equation given by the initial values $u_0$ and $u_1$, and the recurrence relation $u_n = au_{n-1} + bu_{n-2}$ for $n \geq 2$. Define the generating function associated to $\{u_n\}_{n=0}^{\infty}$ by $U(x) = \sum_{n=0}^{\infty} u_n x^n$. The recurrence relation implies that $U(x)(1 - ax - bx^2) = u_0 + (u_1 - au_0)x$ in a neighborhood of

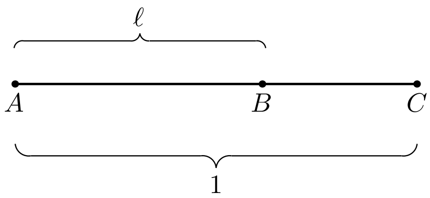


**Figure 2.** Appearance of the golden mean

the origin. If $\alpha$ and $\beta$ denote the roots of the polynomial $p(x) = x^2 - ax - b$, then we may write

$$U(x) = \frac{u_0 + (u_1 - au_0)x}{(1-\alpha x)(1-\beta x)} = \frac{A}{1-\alpha x} + \frac{B}{(1-\beta x)} = A\sum_{n=0}^{\infty} \alpha^n x^n + B\sum_{n=0}^{\infty} \beta^n x^n,$$

where it is an easy matter to solve for $A$ and $B$. Finally, this gives $u_n = A\alpha^n + B\beta^n$. Note that this approach yields a solution to our problem if the roots of $p$ are distinct, namely $\alpha \neq \beta$. A variant of the formula holds if $\alpha = \beta$.

**4.** Using the generating formula for $p(n)$, prove the recurrence formula

$$\begin{aligned} p(n) &= p(n-1) + p(n-2) - p(n-5) - p(n-7) - \cdots \\ &= \sum_{k\neq 0} (-1)^{k+1} p\left(n - \frac{k(3k+1)}{2}\right), \end{aligned}$$

where the right-hand side is the finite sum taken over those $k \in \mathbb{Z}$, $k \neq 0$, with $k(3k+1)/2 \leq n$. Use this formula to calculate $p(5)$, $p(6)$, $p(7)$, $p(8)$, $p(9)$, and $p(10)$; check that $p(10) = 42$.

The next two exercises give elementary results related to the asymptotics of the partition function. More refined statements can be found in Appendix A.

**5.** Let

$$F(x) = \sum_{n=0}^{\infty} p(n)x^n = \prod_{n=1}^{\infty} \frac{1}{1-x^n}$$

be the generating function for the partitions. Show that

$$\log F(x) \sim \frac{\pi^2}{6(1-x)} \qquad \text{as } x \to 1, \text{ with } 0 < x < 1.$$

[Hint: Use $\log F(x) = \sum \log(1/(1-x^n))$ and $\log(1/(1-x^n)) = \sum (1/m)x^{nm}$, so

$$\log F(x) = \sum \frac{1}{m} \frac{x^m}{1-x^m}.$$

Use also $mx^{m-1}(1-x) < 1 - x^m < m(1-x)$.]

**6.** Show as a consequence of Exercise 5 that

$$e^{c_1 n^{1/2}} \leq p(n) \leq e^{c_2 n^{1/2}}$$

for two positive constants $c_1$ and $c_2$.
[Hint: $F(e^{-y}) = \sum p(n)e^{-ny} \leq Ce^{c/y}$ as $y \to 0$. So $p(n)e^{-ny} \leq ce^{c/y}$. Take $y = 1/n^{1/2}$ to get $p(n) \leq c' e^{c' n^{1/2}}$. In the opposite direction

$$\sum_{n=0}^{m} p(n)e^{-ny} \geq C(e^{c/y} - \sum_{n=m+1}^{\infty} e^{cn^{1/2}} e^{-ny}),$$

and it suffices to take $y = Am^{-1/2}$ where $A$ is a large constant, and use the fact that the sequence $p(n)$ is increasing.]

**7.** Use the product formula for $\Theta$ to prove:

(a) The "triangular number" identity

$$\prod_{n=0}^{\infty} (1 + x^n)(1 - x^{2n+2}) = \sum_{n=-\infty}^{\infty} x^{n(n+1)/2},$$

which holds for $|x| < 1$.

(b) The "septagonal number" identity

$$\prod_{n=0}^{\infty} (1 - x^{5n+1})(1 - x^{5n+4})(1 - x^{5n+5}) = \sum_{n=-\infty}^{\infty} (-1)^n x^{n(5n+3)/2},$$

which holds for $|x| < 1$.

**8.** Consider Pythagorean triples $(a, b, c)$ with $a^2 + b^2 = c^2$, and with $a, b, c \in \mathbb{Z}$. Suppose moreover that $a$ and $b$ have no common factors.

(a) Show that either $a$ or $b$ must be odd, and the other even.

(b) Show in this case (assuming $a$ is odd and $b$ even) that there are integers $m, n$ so that $a = m^2 - n^2$, $b = 2mn$, and $c = m^2 + n^2$. [Hint: Note that $b^2 = (c - a)(c + a)$, and prove that $(c - a)/2$ and $(c + a)/2$ are relatively prime integers.]

(c) Conversely, show that whenever $c$ is a sum of two-squares, then there exist integers $a$ and $b$ such that $a^2 + b^2 = c^2$.

**9.** Use the formula for $r_2(n)$ to prove the following:

(a) If $n = p$, where $p$ is a prime of the form $4k + 1$, then $r_2(n) = 8$. This implies that $n$ can be written in a unique way as $n = n_1^2 + n_2^2$, except for the signs and reordering of $n_1$ and $n_2$.

(b) If $n = q^a$, where $q$ is prime of the form $4k + 3$ and $a$ is a positive integer, then $r_2(n) > 0$ if and only if $a$ is even.

(c) In general, $n$ can be represented as the sum of two squares if and only if all the primes of the form $4k+3$ that arise in the prime decomposition of $n$ occur with even exponents.

**10.** Observe the following irregularities of the functions $r_2(n)$ and $r_4(n)$ as $n$ becomes large:

(a) $r_2(n) = 0$ for infinitely many $n$, while $\limsup_{n\to\infty} r_2(n) = \infty$.

(b) $r_4(n) = 24$ for infinitely many $n$ while $\limsup_{n\to\infty} r_4(n)/n = \infty$.

[Hint: For (a) consider $n = 5^k$; for (b) consider alternatively $n = 2^k$, and $n = q^k$ where $q$ is odd and large.]

**11.** Recall from Problem 2 in Chapter 2, that

$$\sum_{n=1}^{\infty} d(n)z^n = \sum_{n=1}^{\infty} \frac{z^n}{1-z^n}, \qquad |z| < 1$$

where $d(n)$ denotes the number of divisors of $n$.

More generally, show that

$$\sum_{n=1}^{\infty} \sigma_\ell(n)z^n = \sum_{n=1}^{\infty} \frac{n^\ell z^n}{1-z^n}, \qquad |z| < 1$$

where $\sigma_\ell(n)$ is the sum of the $\ell^{\text{th}}$ powers of divisors of $n$.

**12.** Here we give another identity involving $\theta^4$, which is equivalent to the four-squares theorem.

(a) Show that for $|q| < 1$

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2}.$$

[Hint: The left-hand side is $\sum \sigma_1(n)q^n$. Use $x/(1-x)^2 = \sum_{n=1}^{\infty} nx^n$.]

(b) Show as a result that

$$\sum_{n=1}^{\infty} \frac{nq^n}{1-q^n} - \sum_{n=1}^{\infty} \frac{4nq^{4n}}{1-q^{4n}} = \sum_{n=1}^{\infty} \frac{q^n}{(1-q^n)^2} - 4\sum_{n=1}^{\infty} \frac{q^{4n}}{(1-q^{4n})^2} = \sum \sigma_1^*(n)q^n$$

where $\sigma_1^*(n)$ is the sum of the divisors of $d$ that are not divisible by 4.

(c) Show that the four-squares theorem is equivalent to the identity

$$\theta(\tau)^4 = 1 + 8\sum_{n=1}^{\infty} \frac{q^n}{(1+(-1)^n q^n)^2}, \qquad q = e^{\pi i \tau}.$$

## 5 Problems

**1.*** Suppose $n$ is of the form $n = 4^a(8k+7)$, where $a$ and $k$ are positive integers. Show that $n$ cannot be written as the sum of three-squares. The converse, that every $n$ that is *not* of that form *can* be written as the sum of three-squares, is a difficult theorem of Legendre and Gauss.

**2.** Let $\mathrm{SL}_2(\mathbb{Z})$ denote the set of $2 \times 2$ matrices with integer entries and determinant 1, that is,

$$\mathrm{SL}_2(\mathbb{Z}) = \left\{ g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \text{ and } ad - bc = 1 \right\}.$$

This group acts on the upper half-plane by the fractional linear transformation $g(\tau) = (a\tau + b)/(c\tau + d)$. Together with this action comes the so-called fundamental domain $\mathcal{F}_1$ in the complex plane defined by

$$\mathcal{F}_1 = \{\tau \in \mathbb{C} : |\tau| \geq 1, \quad |\mathrm{Re}(\tau)| \leq 1/2 \text{ and } |\mathrm{Im}(\tau)| \geq 0\}.$$

It is illustrated in Figure 3.

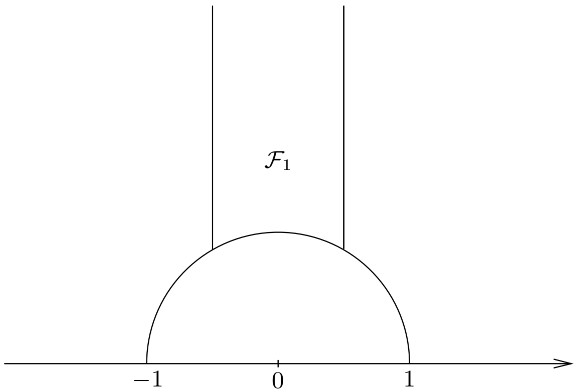


**Figure 3.** The fundamental domain $\mathcal{F}_1$

Consider the two elements in $\mathrm{SL}_2(\mathbb{Z})$ defined by $S(\tau) = -1/\tau$ and $T_1(\tau) = \tau + 1$. These correspond (for example) to the matrices

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},$$

respectively. Let $\mathfrak{g}$ be the subgroup of $\mathrm{SL}_2(\mathbb{Z})$ generated by $S$ and $T_1$.

(a) Show that for every $\tau \in \mathbb{H}$ there exists $g \in \mathfrak{g}$ such that $g(\tau) \in \mathcal{F}_1$.

(b) We say that two points $\tau$ and $\tau'$ are congruent if there exists $g \in \mathrm{SL}_2(\mathbb{Z})$ such that $g(\tau) = w$. Prove that if $\tau, w \in \mathcal{F}_1$ are congruent, then either $\mathrm{Re}(\tau) =$